

Two Derivation Obstructions for Compact Quasinilpotent Generators

Partial-result packet for arXiv:1008.5241

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Status: candidate substantial partial result, likely valid; human review requested. For $A_T = \overline{\text{span}}\{T^n : n \geq 1\}$, weak amenability forces both $A_T = \overline{A_T^2}$ and $[B(H), T] \cap A_T = \{0\}$. These necessary conditions rule out every nonzero compact unilateral weighted shift and every fractional-integration operator $J_{1/r}$, including the classical Volterra operator. They do not decide whether an exceptional compact quasinilpotent satisfying both conditions can be weakly amenable.

1 The source question

For $T \in B(H)$, put

$$A_T = \overline{\text{span}}\{T^n : n \geq 1\}.$$

This is a commutative, possibly nonunital Banach algebra. The source calls T weakly amenable when A_T is weakly amenable. Remark 2.3 asks:

Can a compact quasinilpotent operator be weakly amenable?

The same remark's question about compact normal operators is answered later in the source, but the quasinilpotent question is left open [1].

Hence, T is not similar to a normal operator.

Remark 2.3. Theorem 2.2 shows that there exists a compact triangular operator with infinite spectrum which is a weakly amenable operator but is not similar to a normal operator. However, we do not know if a compact quasinilpotent operator can be weakly amenable. It would be very interesting to know whether this result is true. Indeed, if any compact quasinilpotent operator can not be weakly amenable, then it is easy to get that a compact operator T with finite spectrum is weakly amenable if and only if T is similar to a normal operator.

3. COMPACT TRIANGULAR CHARACTER AMENABLE OPERATOR

Let Q be the Volterra operator on infinite-dimension Hilbert space, then by [8] Corollary 2.7 and [2] Corollary 5.11, Q is a compact quasinilpotent operator which is character amenable. However, the lattice of invariant subspaces of Q is a continuous nest. i.e. Q is not a triangular operator. In this section, we will prove that $T = \begin{pmatrix} 0 & N^{\frac{1}{2}} \\ 0 & N \end{pmatrix}$ is a character amenable operator, for any normal operator N with spectrum σ . Hence, there exists a compact triangular operator which is a character amenable operator but is not similar to a normal operator.

In [8], Sangani-Monfared obtained a necessary and sufficient condition for a Banach algebra

Figure 1: Source Remark 2.3 and its Volterra discussion (PDF page 4).

For commutative Banach algebras we use the standard characterization quoted by the source: weak amenability is equivalent to the vanishing of every bounded derivation into every symmetric Banach bimodule. This formulation makes two elementary but effective obstructions visible.

2 The two obstructions

Theorem 1 (Power-span and commutator obstructions). *Let $0 \neq T \in B(H)$. If either*

1. $T \notin \overline{\text{span}}\{T^n : n \geq 2\}$, or
2. there is $X \in B(H)$ such that $0 \neq [X, T] \in A_T$,

then A_T is not weakly amenable. Consequently, every possible positive answer to the source question must satisfy

$$A_T = \overline{A_T^2}, \quad [B(H), T] \cap A_T = \{0\}. \quad (1)$$

Proof. First observe that

$$\overline{A_T^2} = \overline{\text{span}}\{T^n : n \geq 2\}. \quad (2)$$

Under condition 1, Hahn–Banach gives a bounded functional $\delta : A_T \rightarrow \mathbb{C}$ such that $\delta(T) = 1$ and $\delta(\overline{A_T^2}) = 0$. Make $\mathbb{C}$ a symmetric A_T -bimodule with the trivial actions. Then for $a, b \in A_T$,

$$\delta(ab) = 0 = a \cdot \delta(b) + \delta(a) \cdot b.$$

Thus δ is a nonzero bounded derivation into a symmetric module, so A_T is not weakly amenable.

Under condition 2, write $q = [X, T]$. Since $q \in A_T$, it commutes with T . Hence, for every polynomial p with $p(0) = 0$,

$$[X, p(T)] = p'(T)q \in A_T. \quad (3)$$

The commutator map is bounded on $B(H)$, so density and the closedness of A_T extend it to a bounded derivation $D : A_T \rightarrow A_T$, $D(a) = [X, a]$. The target A_T , with its usual multiplication actions, is symmetric; moreover $D(T) = q \neq 0$. Again weak amenability fails. Finally, (2) turns the negation of condition 1 into the first assertion of (1), and the negation of condition 2 into the second. $\square$

Proof intuition. Weak amenability forbids every continuous first-order differential on A_T . If the first power survives modulo products, a scalar functional detecting it is already such a differential. If an ambient commutator lands back in the power algebra, the usual operator commutator restricts to a differential on A_T . A positive example must evade both mechanisms.

3 Homogeneous generators and weighted shifts

Proposition 2 (Gauge obstruction). *Suppose A_T admits a strongly continuous uniformly bounded circle action $(\gamma_\zeta)_{|\zeta|=1}$ satisfying $\gamma_\zeta(T) = \zeta T$. Then A_T is not weakly amenable.*

Proof. The first Fourier projection

$$P_1(a) = \int_{\mathbb{T}} \zeta^{-1} \gamma_\zeta(a) dm(\zeta)$$

is bounded. It satisfies $P_1(T) = T$ and $P_1(T^n) = 0$ for $n \geq 2$. Therefore T cannot belong to the closed span of the higher powers, and Theorem 1(1) applies. $\square$

Corollary 3 (Compact unilateral weighted shifts). *Every nonzero compact unilateral weighted shift is quasinilpotent and is not weakly amenable.*

Proof. Let $We_n = w_n e_{n+1}$ on $\ell^2(\mathbb{N}_0)$. Compactness is equivalent to $w_n \rightarrow 0$. Given $\varepsilon > 0$, all but finitely many factors in every long consecutive product of weights have modulus at most ε ; hence $\limsup_k \|W^k\|^{1/k} \leq \varepsilon$, and W is quasinilpotent.

For $|\zeta| = 1$, define $U_\zeta e_n = \zeta^n e_n$. Then $U_\zeta W U_\zeta^* = \zeta W$. Conjugation gives the circle action in Proposition 2; norm continuity on A_T follows first for polynomials and then by uniform approximation. Thus W is not weakly amenable. $\square$

4 Fractional integration, including Volterra

For $\alpha > 0$, let J_α be the Riemann–Liouville operator on $L^2(0, 1)$:

$$(J_\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt. \quad (4)$$

The case $\alpha = 1$ is the classical Volterra operator.

Theorem 4 (A fractional Volterra family). *For every integer $r \geq 1$, the compact quasinilpotent operator $J_{1/r}$ is not weakly amenable.*

Proof. We first record compactness and quasinilpotence. Truncate the convolution kernel $s^{\alpha-1}/\Gamma(\alpha)$ to $s \geq \varepsilon$. The resulting kernel on $(0, 1)^2$ is square integrable, so its operator is Hilbert–Schmidt. Young’s inequality bounds the discarded operator by

$$\frac{1}{\Gamma(\alpha)} \int_0^\varepsilon s^{\alpha-1} ds = \frac{\varepsilon^\alpha}{\Gamma(\alpha+1)}.$$

Thus J_α is a norm limit of compact operators.

The beta-integral identity gives the semigroup law $J_\alpha J_\beta = J_{\alpha+\beta}$. Young's inequality also gives

$$\|J_\alpha^n\| = \|J_{n\alpha}\| \leq \frac{1}{\Gamma(n\alpha + 1)}.$$

Stirling's formula now yields $\lim_n \|J_\alpha^n\|^{1/n} = 0$.

Let M_x denote multiplication by x . A direct kernel calculation from (4) gives

$$[M_x, J_\alpha] = \alpha J_{\alpha+1}. \quad (5)$$

If $\alpha = 1/r$, then the semigroup law turns (5) into

$$[M_x, J_{1/r}] = \frac{1}{r} J_{1/r}^{r+1} \in A_{J_{1/r}} \setminus \{0\}.$$

Theorem 1(2) completes the proof. □

In particular, J_1 is not weakly amenable. The source records the Volterra operator as character amenable; the conclusion here is consistent with that statement and shows that character amenability is strictly weaker for this canonical compact quasinilpotent.

5 Boundary of the result and upgrade audit

Theorem 1 gives necessary, not sufficient, conditions. A full negative solution would follow from a universal theorem that every nonzero compact quasinilpotent either fails $A_T = \overline{A_T^2}$ or admits a nonzero commutator $[X, T] \in A_T$. Neither step is presently justified. Volterra-type power algebras show that the first alternative need not be automatic, while the second asks for exact membership in the range of the Rosenblum operator $X \mapsto [X, T]$, not merely approximate triangular information.

Eight focused upgrade attempts covered the current literature, nilpotent and finite-type reductions, transfer of the abstract radical weakly amenable example, normalized powers, the essentiality obstruction, gauge actions, commutator derivations, and the universal Rosenblum/triangularization route. The last two successful mechanisms produce the two broad classes above; the remaining universal step has no credible closure argument. The full source question therefore remains open in this packet.

Verification and novelty. The proof is self-contained apart from the standard symmetric-module characterization of weak amenability quoted by the source. Human review should focus on the range assertion (3), the norm-continuity of the gauge action, and the kernel calculation (5). On 12 August 2026, all cheap run indexes and exact/current literature searches were checked. Later work on singly generated radical operator algebras treats compact weighted shifts and Volterra-type algebras but does not decide weak amenability [3]. An earlier abstract construction gives a singly generated radical weakly amenable Banach algebra, but on a Banach space isometric to ℓ^1 , not as the Hilbert-space compact-operator example asked for here [2]. Novelty confidence is moderate and provisional.

Human-review recommendation. Review as a substantial partial result. It materially narrows any possible positive example and settles two natural, infinite families, but it is not a proof or counterexample to the existence question.

References

- [1] L. Y. Shi, Y. J. Wu, and Y. Q. Ji, *An example of weakly amenable and character amenable operator*, arXiv:1008.5241; Illinois J. Math. 55 (2011), no. 4, 1415–1422.
- [2] R. J. Loy, C. J. Read, V. Runde, and G. A. Willis, *Amenable and weakly amenable Banach algebras with compact multiplication*, J. Funct. Anal. 171 (2000), 78–114.
- [3] J. Peters, *Singly generated radical operator algebras*, arXiv:2309.13530; New York J. Math. 30 (2024), 398–421.